

# EXPERIMENTAL STUDY OF THE DEPENDENCE OF THERMODYNAMIC PROPERTIES OF THE WATER + PROPANOL-1 BINARY SYSTEM ON ITS COMPONENT COMPOSITION OVER A WIDE RANGE OF STATE PARAMETERS

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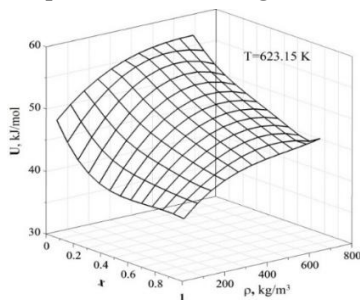
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Measurements of the  $(p, T-p, \rho, T)_x$  - dependences for 1-propanol aqueous solutions with compositions of 1-propanol ( $x = 0.1, 0.2, 0.3, 0.5, 0.8, 0.9$  molar fractions), in two-phase, one-phase (liquid and vapor), and supercritical fluid states over temperatures ranging from 373.15 K to 673.15 K, densities between 1.5 kg/m<sup>3</sup> and 850 kg/m<sup>3</sup>, and pressures up to 50 MPa have been carried out. The experimental values of  $(p, T-p, \rho, T)_x$ -dependencies are described by a thermal equation of state of virial form — an expansion of the compressibility factor  $Z = p/(RT\rho_m)$  into series of reduced density and temperature powers [1].

$$Z = p/RT\rho_m = 1 + \sum_{i=1}^m \sum_{j=0}^{n_i} a_{ij} \omega^i / \tau^j, \text{ and } p = RT\rho_m \left[ 1 + \sum_{i=1}^m \sum_{j=0}^{n_i} a_{ij} \omega^i / \tau^j \right]$$

Herein:  $\rho_m$  is the molar density (mol/m<sup>3</sup>),  $\omega = \rho/\rho_c$  and  $\tau = T/T_c$  are the expressions for reduced density and reduced temperature respectively, where  $\rho_c$  and  $T_c$  represent critical density and critical temperature;  $R = 8.314$  J/(mol·K) is a universal (molar) gas constant. The average relative deviation of calculated pressure values using this equation compared to experimental data amounts to 1.2%.

Based on Equation [1] and thermodynamic relations [2], variations of the main thermodynamic properties of the studied solutions were computed. A plot illustrating the dependence of internal energy changes ( $U$ ) on solution composition ( $x$ ) and density ( $\rho$ ) at a temperature of 623.15 K is presented in the figure.



1. Osmanova B.K., Bazaev A.R., Bazaev E.A. Experimental research of thermodynamic properties of water+ aliphatic alcohol mixtures in the wide parameters of state // Journal of Physics: Conf.Series 1385 (2019).

2. Shpilrain E.E., Kesselman P.M. Fundamentals of Theory of Thermophysical Properties of Substances. Moscow: Energiia Publ., 1977. 248 pp.